

Ernie Wang

469-961-3315 | erniewang2025@u.northwestern.edu | erniewang.com | github.com/erniewang

EDUCATION

Northwestern University

B.A. in Computer Science, B.M. in Jazz Studies

Evanston, IL

Jun. 2025

WORK AND PROJECT EXPERIENCE

Computer Vision Developer | *Node.js, Flask, Python, p5.js, PoseNet, VampNet, Socket.io*

Mar. 2025 – Jun. 2025

- Implemented an interactive browser application that captures and analyzes body movement using PoseNet (p5.js) and generates responsive music in real time via the VampNet generative model.
- Engineered a dual-service architecture integrating Node.js (data ingestion) and Flask (Python processing) communicating via Socket.io.
- Implemented real-time tempo estimation by applying FFT on pose-net velocity data to synchronize musical output (pitch, playback rate) with performer motion.
- Built adaptive interface controls for adjusting normalization, frame rate, and sensitivity parameters to calibrate system response to performer behavior.

Harmonizer | *Python, FastAPI, JavaScript, MusicXML, JSON*

Nov. 2024 – Present

- Built a rule-based harmonization engine that transforms MusicXML melodies into solis using structured pitch representations mapped to predefined multi-voice voicing templates.
- Enforced constraint validation for key-signature consistency, valid monophonic MusicXML input, and melody-chord pairs prior to emitting valid MusicXML output.
- Improved runtime by 9× via two-tier caching: in-process chord-note maps and post-process parse results persisted in localStorage to eliminate redundant resolution.
- Designed deterministic fallback resolution logic to handle non-chord tones and any unresolved mappings, ensuring complete melodic coverage without invalid XML or unwritten chords.
- Architecting a cloud-based collaboration platform enabling users to publish and share harmonization formulas and generated arrangements.

Expedition Orange | *Node.js, Express, PostgreSQL, Vercel, Supabase, Render, Prisma*

Feb. 2026 – March 2026

- Reduced a full-stack codebase by approximately 40%, removing unused authentication flows, dead admin routes, and database models that added complexity without business value.
- Cut monthly infrastructure costs from hundreds of dollars to zero by migrating to free-tier managed services without disrupting core application functionality.
- Designed and built a client-facing admin interface, allowing non-technical users to manage key database records directly via the browser.

Composition Engine (CLI) | *Python, Music21, MusicXML, ABC*

Nov. 2025 – Dec. 2025

- Built an experimental CLI pipeline that converts MusicXML melodies to ABC notation as an intermediate representation and back, enabling model-assisted harmonization and continuation experiments.
- Designed a file-backed transformation history mechanism that records intermediate pipeline states, with validation enforcing rhythmic alignment and key constraints and automatic regeneration of structurally invalid outputs.
- Developed targeted test cases for ABC cut/paste operations and deterministic XML round-trip validation to support future AI-driven composition workflows.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, C++, Racket, Java, Go

Frameworks/Tools: Git, Linux, SQL (PostgreSQL, SQLite), FastAPI, Flask, Pydantic, Pandas, NumPy, Node.js, AWS, React, Next.js, Electron, TailwindCSS, Vite, BeautifulSoup, Puppeteer, XML

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Databases, Security, Machine Learning/AI, Operating Systems, Distributed Systems

OTHER EXPERIENCE

Musician (Self-Employed)

Sep. 2020 – Present

- Performed with Northwestern University Jazz Orchestra (2020–2024) and in a variety of university bands across multiple musical styles, including live and virtual performances.
- Utilized AI music generation tools (Musicfy, Udio) alongside music production software (MuseScore, Finale, Logic Pro X, Audacity) and custom-programmed tools to assist in composition and production workflows.
- Conducted private lessons, teaching saxophone and clarinet fundamentals and improvisational listening skills to students across a wide range of experience levels.